
Proof of Frank-Wolfe Method

To solve the constrained optimization $\min_{x \in X} f(x)$, consider the Frank-Wolfe (Conditional Gradient) method:

$$\begin{aligned} x^{r+1} &= x^r + \alpha_r(\bar{x}^r - x^r) \\ \bar{x}^r &= \arg \min_{x \in X} \nabla f(x^r)^\top (x - x^r) \end{aligned} \quad (1)$$

Now we provide a convergence rate analysis. We have to assume that f is convex and L Lipschitz-smooth (i.e. ∇f is L -Lipschitz continuous), also assume that X is compact. Based on these two assumptions, we can further assume that there exists $c > 0$ such that $\|x\|^2 \leq c$ and $f(x) - f(x^*) \leq Lc/2$ for all $x \in X$. Here we denote $x^* \in X$ the minimizer.

Now that

$$\begin{aligned} f(x^{r+1}) &= f(x^r + \alpha_r(\bar{x}^r - x^r)) \\ (L\text{-smooth}) &\leq f(x^r) + \alpha_r \langle \nabla f(x^r), \bar{x}^r - x^r \rangle + \frac{L\alpha_r^2}{2} \|\bar{x}^r - x^r\|^2 \\ (\text{Think about why?}) &\leq f(x^r) + \alpha_r \langle \nabla f(x^r), x^* - x^r \rangle + cL\alpha_r^2 \\ &= (1 - \alpha_r)f(x^r) + \alpha_r(f(x^r) + \langle \nabla f(x^r), x^* - x^r \rangle) + cL\alpha_r^2 \\ (\text{Convexity}) &\leq (1 - \alpha_r)f(x^r) + \alpha_r f(x^*) + cL\alpha_r^2 \end{aligned} \quad (2)$$

we have (by re-arranging terms)

$$f(x^{r+1}) - f(x^*) \leq (1 - \alpha_r)(f(x^r) - f(x^*)) + cL\alpha_r^2. \quad (3)$$

Denote $e^r = f(x^r) - f(x^*)$, we have

$$e^{r+1} \leq (1 - \alpha_r)e^r + cL\alpha_r^2. \quad (4)$$

Now if we take $\alpha_r = 2/(r+2)$, using mathematical induction (try it! proof is right below) we have the conclusion

$$e^r \leq \frac{Lc}{r+2}. \quad (5)$$

This is readily the rate of convergence that we are looking for.

Now we show how to get (5) from (4) using induction. For $r = 0$, by Lipschitz-smooth we have $e^0 = f(x^0) - f(x^*) \leq Lc/2$ (by our assumption). Now suppose $e^r \leq \frac{Lc}{r+2}$ holds, then

$$e^{r+1} \leq (1 - \alpha_r)e^r + cL\alpha_r^2 \leq (1 - \alpha_r)\frac{Lc}{r+2} + cL\alpha_r^2 = \frac{(r+1)Lc}{(r+2)^2} \leq \frac{Lc}{r+3}$$

where the last inequality is due to $\frac{(r+1)}{(r+2)^2} \leq \frac{1}{r+3}$ for any natural number r .